Samba Gangineni

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Atlanta, GA

Profile

Senior Software Engineer with 5+ years of experience architecting and delivering scalable backend systems, distributed data pipelines, and production-grade AI applications. Strong track record building microservices (Java Spring Boot, Python FastAPI, Node.js), ETL/training pipelines (Apache Spark/Scala), and AI-powered customer experiences including LLM chatbots and recommendation platforms with observability, reliability, and cost controls across cloud environments (AWS, GCP).

Skills

- **Programming:** Java, Kotlin, Python (FastAPI, Flask), JavaScript/Node.js, Go, SQL, Bash, HTML, CSS, HCL
- **Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Helm, Terraform, Pulumi, Git, Jenkins, CI/CD, Redis
- **Data & AI Systems:** Apache Spark, Scala, Temporal, RabbitMQ, Elasticsearch, Vector Search (e.g., FAISS/Weaviate), LangChain, LangGraph, Google ADK, LangSmith (observability), TensorFlow
- **Databases:** MongoDB, MSSQL Server

Experience

The Home Depot *Jun 2025 – Present — Atlanta, GA*
Senior Software Engineer

- Built and enhanced the **Magic Apron Assistant** (LLM chatbot) using agentic orchestration frameworks (e.g., Google ADK, LangChain/LangGraph), improving response quality and reliability through structured tool calling and workflow routing.
- Implemented efficient **conversation memory** and context management patterns for multi-turn interactions, enabling scalable session handling and improved user experience.
- Added end-to-end **observability** for AI workflows (e.g., tracing/metrics/logging with LangSmith-style telemetry patterns) to monitor latency, errors, and LLM usage and improve production troubleshooting.
- Implemented **rate limiting and backpressure** controls to protect downstream model/tool dependencies and improve system stability under load.
- Collaborated with Google on a specialized **Garden Agent** solution for garden-related customer experiences, leveraging Magic Apron capabilities and integrating domain-specific tools and flows.
- Led the design and prototype development for **unifying Live Chat and Magic Apron**, enabling routing across multiple AI agents and laying the foundation to scale additional agent connections in the future.
- Designed and developed an internal engineering tool with both **TUI and Web UI** interfaces to help engineers harness AI for faster design and implementation workflows.
- Contributed to **model deployment**, **ETL/data pipelines**, and **recommendations serving** capabilities, supporting batch/online workflows and operational metrics.

Availity *Feb 2020 – Jun 2025 — Atlanta, GA*
Software Engineer IV

- Led a redesign of a legacy application by integrating Apache Spark pipelines and Scala, enhancing performance & scalability and reducing cost by 90% to support multi-tenant workloads.
- Developed an internal Confluence chatbot for knowledge management—implemented REST endpoints using Python FastAPI, integrated LLMs with Retrieval-Augmented Generation (RAG), and leveraged vector

search for efficient retrieval and relevance ranking.

- Developed an AI agent to generate documentation from function signatures/metadata by constructing prompts and using Amazon Q integrated with LangChain to produce human-readable documentation.
- Designed and implemented a stateless product pipeline with event-driven architecture using Temporal, reducing tight coupling and improving reliability.
- Transitioned infrastructure code and customer environments from Pulumi to Terraform, improving provisioning repeatability and deployment stability.
- Achieved a 5x performance improvement in data deduplication by redesigning the architecture with Kotlin coroutines to minimize database and RabbitMQ load.
- Streamlined data ingestion using Temporal and managed CI/CD deployments into EKS with Terraform.
- Spearheaded MongoDB sharding and large-scale dataset regeneration, supporting high-volume processing workloads.
- Managed large-scale ingestion pipelines using Spark, AWS Lambda, EMR, and Kubernetes for multi-tenant processing at scale.
- Built an inference/analytics model to detect gaps in structured data using association mining techniques.
- Automated terminology updates with Python and SQL Server stored procedures to support normalization and downstream analytics.

Diameter Health *Jun 2019 – Aug 2019 — Boston, MA*
Informatics and Engineering Intern

- Developed a transpiler in Golang to convert Clinical Quality Language into JavaScript, automating boilerplate code generation.
- Designed and implemented a synthetic data generator for clinical HL7 messages in JavaScript, accelerating testing and development cycles.

Research

Synthetic Health Data Generation *Jan 2026*
TE-CTABGAN+: A Unified GAN Framework for Generating Synthetic Electronic Health Records

- Currently researching methods to generate synthetic healthcare data using Mimic IV and Synthea sources.
- Compared various GAN and Diffusion models, and finalized a GAN architecture that incorporates temporal aspects for realistic data simulation.
- Publication: https://doi.org/10.1007/978-3-032-15632-7_20

ACM DEBS *Jun 2019*
Distributed and Event Based Systems

- Architected a real-time object detection system for LiDAR data using clustering techniques and convolutional neural networks.
- Publication: https://doi.org/10.1145/3328905.3330297

Education

Boston University *MSc in Computer Info Systems* *Grad. Jan 2020*

- GPA: 3.86/4.0
- **Coursework:** Big Data Analytics with Spark, Artificial Intelligence, DataMining & Business Intelligence

Awards

2019 ACM DEBS Grand Challenge Winner (Object Recognition)
2018 1st Place in BigRedHacks (ResQu - An SOS Application)
2018 3rd Place in Att Hacks (Imound - An App for the Visually Challenged)